Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/17/2005
PI Name:	Rydberg, Bjorn Ph.D.		
Project Title:	DNA Double-Strand Break Induction by HZE Particles. Repair & Consequences in Human & Rodent Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	PI is deceased, as of December 2011.		
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	08/01/2002	**End Date:**	07/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Pluth, Janice (Lawrence Berkeley National Laboratory)		
Grant/Contract No.:	T-9493W		
Performance Goal No.:			
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	A concern for long-term deep space travel is the possibility that galactic cosmic rays with high charge and energy (HZE particles) could cause deleterious effects from mechanisms that are fundamentally different from high LET particles of lower energy, which we have a much better knowledge of. Despite this concern, it remains likely that the mechanisms of damage are similar and differ only slightly due to the spatial distribution of induced damage, which on the cellular level is reflected in the complexity and distribution of double-strand breaks in DNA (DSBs). The experiments described in this proposal are designed to support the hypothesis that predictable DSBs cause the major cytogenetic and killing effects of HZE particle exposure. To test this hypothesis, we will first determine the initial yield of DSBs from HZE Fe ions at low doses (0.1- 10 Gy) to test for linearity with dose, determine the presence of heat-labile sites that can convert into DSBs, and measure the distribution of DSBs along megabase-pair stretches of DNA. Secondly, we will test the possibility that DSBs are induced not only in irradiated cells but also in non-irradiated bystander cells. Thirdly, we will measure the proportion of DSBs that do not rejoin 24 hrs after Fe HZE ion irradiation in the low-dose region and		

Task Description:	compare such non-rejoined DSBs with the number of breaks in premature condensed chromosomes (PCC breaks) remaining after 24 hrs. Fourth, we will use mutant cell lines defective in non-homologous end-joining and homologous recombination to test the involvement of DSB repair in cell killing, comparing HZE particles with low-energy high LET radiation. These experiments will be performed with varying energies of Fe ions to determine RBE/LET relationships, and also with secondary particles generated in shielding materials. 1. To measure DNA double-strand break induction by HZE Fe-ions, low energy high LET N-ions and X-rays in the dose-range of 0.1 – 10 Gy to look for specific low-dose effects. 2. To test the possibility that DSBs can be induced in non-irradiated bystander cells. The medium transfer technique is used to test for clastogenic factors in the growth medium of cells irradiated with HZE Fe ions. 3. To measure in human cells the number of DSBs remaining unrejoined 24 hours after irradiation and compare this with remaining unrejoined premature condensed chromosomes (PCC) in the same dose and time interval. 4. To compare micronuclei formation induced by HZE Fe ions of various energies in normal cells and cells defective in non-homologous end-joining. It is important for risk estimation to verify that there are no unexpected effects from HZE ions. Also, mechanistic understanding of bystander effects may help to design countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Low dose radiation data is of general interest for prediction of health effects.
Task Progress:	Using a newly developed dual-label pulsed-field gel electrophoresis method, we have measured the initial induction of DSBs in human HeLa cells and Ku80 mutated Chinese hamster xrs6 cells for X-rays, N-ions and 1 GeV/n Fe ions in the dose range 0.2 – 2 Gy. We find that DSB induction from Fe ions falls between X-rays and N-ions, as was expected theoretically. This indicates that there are no significant unusual mechanisms for induction of DSBs by the HZE particles. We have also measured chromatid breaks in prematurely condensed chromosomes (PCC breaks) and found that there is about 1 PCC break per 6 DSBs for both 1 GeV/n Fe-ions and low energy N-ions. This again indicates no unusual mechanisms for induction of cytogenetic damage. On the other hand, we have found un unexpected large difference in micro-nuclei induction from 600 MeV/n Fe ions compared to 1 GeV/n Fe ions, which is not understood. We have also performed bystander experiments with N-ions and Fe ions and detected low levels of DSBs and breaks in prematurely condensed chromosomes (PCC breaks) in bystander cells. However, these results have been difficult to reproduce, and all data together at present lack statistical significance.
Bibliography Type:	Description: (Last Updated: 01/17/2005)
Abstracts for Journals and Proceedings	Rydberg, B. "Low-dose measurement of DSBs using a dual-label FAR assay" The 14th Annual Space Radiation Health Investigators' Workshop Huston , Apr-2003
Abstracts for Journals and Proceedings	Rydberg, B.; Cooper, B.; Groesser, T. "Relationship between DNA double-strand breaks and chromosome breaks in irradiated and bystander cells" The 15th Annual Space Radiation Health Investigators' Workshop Port Jefferson , May-2004
Abstracts for Journals and Proceedings	Pluth, J.; Groesser, T.; Cooper, B.; Rydberg, B. "Does HZE radiation-induced DNA damage differ from low-energy high LET radiation damage?" Bioastronautics Investigators' Workshop Galveston , Jan-2005